

LE HONG PHAT

Data Engineer

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SUMMARY

Final-year Information Technology student with 9 months of hands-on experience as a Data Engineer Intern at AFFINA Insurance, specializing in building real-time CDC pipelines, batch transformation, and data platform governance. Proficient in architecting scalable streaming architectures using Kafka and ClickHouse, as well as orchestrating dbt pipelines with Airflow. Passionate about implementing automated data cataloging, lineage tracking, and SLO monitoring to ensure enterprise-grade data quality and system reliability.

TECHNICAL SKILLS

- **Languages:** Python, SQL, Java
- **Data Pipeline & Orchestration:** Airflow, Kafka, Spark, Debezium, dbt, Trino, ETL/ELT
- **Databases:** PostgreSQL, Oracle, MySQL, MongoDB, ClickHouse, Redis
- **Infrastructure & Monitoring:** Docker, Prometheus, Grafana, Linux, Git
- **Cloud Exposure:** AWS, GCP
- **Concepts:** CDC, ETL/ELT, Data Modeling, Stream/Batch Processing, Data Lake/Warehouse

WORK EXPERIENCE

Data Engineer Intern

Sep 2025 – May 2026

AFFINA Insurance

- **Data Platform Architecture (Phase 1):** Built the end-to-end data platform capturing real-time MySQL CDC events via Debezium and scheduled Excel data, consolidating them into staging tables and normalized marts with ingestion latency under 2 seconds.
- **Enterprise Data Consolidation (Phase 2):** Evolved the platform to resolve complex online-offline data integration issues. Implemented a custom Contract Pre-Processing and Policy Parser to standardize schema discrepancies across 50+ contract variants, enabling consistent policy reporting.
- **Idempotency & Deduplication:** Designed a real-time deduplication component utilizing Redis Contract Caching to track and validate record uniqueness, ensuring zero data loss and exactly-once insertion into the Operational Data Store.
- **Event-Driven Architecture:** Designed RabbitMQ event routing topology for 5 downstream consumer applications, implementing at-least-once delivery semantics and decoupling the core data platform from business-layer consumers – enabling independent scaling of each processing pipeline.

PROJECTS

FMCG Real-Time Analytics Platform

Jun 2026

github.com/phantle224/fmcg-real-time-analytics

- Architected a dual-path (hot/cold) FMCG analytical platform processing 1,000 POS transactions/second using Apache Kafka as the message broker, ClickHouse Kafka Engine for real-time OLAP ingestion, and Kafka Connect S3 Sink for historical archival.
- Designed ClickHouse MergeTree schema with optimized ORDER BY index keys and monthly partitioning, implementing 2 Materialized Views for pre-aggregation, reducing OLAP query latency from 12s (PostgreSQL) to sub-100ms on 10M+ records.
- Deployed Trino as a federated query engine with dual catalogs (ClickHouse JDBC + Hive Metastore), enabling single-SQL joins across real-time and historical Parquet datasets on MinIO without custom ETL scripts.
- Implemented Cube.js semantic layer to define standardized business metrics (Revenue, Units Sold, Avg Basket) with caching, exposing a PostgreSQL-compatible interface consumed by Grafana dashboards.

Tech: *Python • FastAPI • Apache Kafka • ClickHouse • MinIO • Apache Iceberg • Trino • Cube.js • Grafana • Docker*

Data Platform Governance & Observability Stack
github.com/phatle224/data-platform-governance

Jul 2026

- Deployed OpenMetadata as a centralized data catalog connecting to PostgreSQL, ClickHouse, and Trino from the FMCG Analytics Platform via shared Docker network, automating schema discovery and metadata ingestion across 3 data sources.
- Built a dbt transformation pipeline (Staging → Mart) orchestrated by Apache Airflow, with OpenLineage integration that automatically emits lineage events to OpenMetadata — producing a visual end-to-end data lineage graph from raw source to aggregated mart tables.
- Implemented SLO-based pipeline alerting using Prometheus + StatsD Exporter + AlertManager, defining 4 alert rules (DAG failure, SLO breach > 5 min, scheduler down, high task failure rate) with Slack webhook notifications triggering within 30 seconds.
- Centralized log aggregation from 14+ Docker containers across both projects using Grafana Loki + Promtail with auto-discovery, building a 9-panel Grafana dashboard for real-time error monitoring via LogQL queries.

Tech: *OpenMetadata • OpenLineage • Apache Airflow • dbt • Prometheus • AlertManager • Grafana Loki • Docker*

EDUCATION

Bachelor of Information Technology

Sep 2022 – Expected Mar 2027

Saigon University (SGU), Ho Chi Minh City

- Academic Excellence Scholarship – 3 consecutive semesters (HK1 2025 – HK1 2026). GPA: 7.4/10.
- Relevant coursework: Database Systems, Data Structures & Algorithms, Big Data Technologies, Software Engineering.

CERTIFICATIONS

IBM Data Engineering Professional Certificate - Coursera / IBM

In Progress

IBM Data Analyst Professional Certificate - Coursera / IBM

Feb 2025

SQL (Advanced) Certificate – HackerRank

Dec 2025

LANGUAGES

- **Vietnamese** Native
- **English** B2 (IELTS in progress)